

# Onyx — FileKit reference

Onyx · multi-process OS on Raspberry Pi 4

The reference of **FileKit** (`SD:/lib/filekit.so`, `user/Kits/filekit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: [The Kits](#).

## Contents

1. What it is
2. Using it
3. Index
4. `filekit/filekit.h`
5. `filekit/fsutil.h`
6. `filekit/kvtext.h`

## What it is

FileKit is files and folders: whole files read and written, trees copied, moved and removed, paths, compression (zlib), archives — ZIP read and written, tar / tar.gz / gzip read — and sectioned key / value text documents read and written back (a progress file, a level pack).

|         |                                                                                                                          |
|---------|--------------------------------------------------------------------------------------------------------------------------|
|         |                                                                                                                          |
| Include | <code>#include "filekit/filekit.h"</code>                                                                                |
| Link    | <code>lib/filekit.imp.a</code>                                                                                           |
| Library | <code>SD:/lib/filekit.so</code> — 99 entries in its table<br>( <code>user/Kits/filekit/filekit.abi</code> , append-only) |
| Sources | <code>user/Kits/filekit/</code>                                                                                          |

## Using it

**A whole file, read and written:**

```
#include "filekit/filekit.h"

void *data; unsigned n;
if (fk_load ("SD:/docs/notes.txt", &data, &n) == 0)
{
    /* data: n bytes, a 0 after them */
    fk_save ("SD:/docs/notes.bak", data, n);
    fk_free (data);
}
```

**Folders and paths:**

```
#include "filekit/filekit.h"

char path[FS_PATHL];
```

```

fs_join (path, sizeof path, "SD:/docs", "report.txt");
if (!fs_exists (path)) { /* ... */ }

fk_mkdirs ("SD:/backup/2026/docs");           // every folder of the path
fk_copy ("SD:/docs", "SD:/backup/2026/docs", 0, 0); // a file, or a folder and all
it holds
fk_remove ("SD:/tmp/work");                   // the same, removed

```

### A ZIP archive, read:

```

char err[80];
fk_zip *z = fk_zip_open ("SD:/downloads/pack.zip", err, sizeof err);
if (z)
{
    struct fk_zip_entry e;
    for (int i = 0; i < fk_zip_count (z); i++)
        if (fk_zip_entry (z, i, &e) ax_putln (e.name);
    fk_zip_extract_all (z, 0, "SD:/downloads/pack", 0, 0);
    fk_zip_close (z);
}

```

### A ZIP archive, written:

```

fk_zipw *w = fk_zipw_create ("SD:/backup/docs.zip");
fk_zipw_add (w, "SD:/docs", 0);           // the folder and all it holds
fk_zipw_close (w, 0, 0, err, sizeof err);

```

Other archive formats (tar, tar.gz, gzip) are read through the same calls (fk\_arc\_\*); a program asks the library which formats it handles (fk\_arc\_formats) instead of keeping its own list.

**A settings or progress file with sections, read and written back** (filekit/kvtext.h, C and C++): the entries stay in the file's order, the keys a program does not know are kept, values may hold new lines.

```

fk_kv *kv = fk_kv_load ("SD:/apps/game.app/progress.ini", FK_KV_ESCAPES);
if (!kv) kv = fk_kv_new (FK_KV_ESCAPES);           // no file yet: a fresh
start
int stars = atoi (fk_kv_get (kv, "Player", "level1", "0"));
fk_kv_set (kv, "Player", "level1", "3");
fk_kv_set (kv, "", "last", "level1");             // "": before the first
[header]
fk_kv_save (kv, "SD:/apps/game.app/progress.ini", "# my game's progress");
fk_kv_free (kv);

```

A level pack (FK\_KV\_PIPES): many [level] blocks (fk\_kv\_block, fk\_kv\_blocks, fk\_kv\_block\_name), a value going on over the | lines that follow, and the line of each value (fk\_kv\_line) for an error message. Circuits reads its packs and its progress.ini so.

Blocks of the same name written (the Clock's [alarm] list; fk\_kv\_set reaches only a name's first block): a new block made at the end, its keys set and read by its number —

```

fk_kv *kv = fk_kv_new (0);
int b = fk_kv_block_new (kv, "alarm");           // its number (1-based),
-1 no memory
fk_kv_block_set (kv, b, "time", "07:00");       // replaced in block b,
else added at its end
const char *t = fk_kv_block_get (kv, b, "time", ""); // block b's value (" if
none)
fk_kv_save (kv, "SD:/apps/clock.app/alarms.txt", "# Clock -- the alarms");

```

## Index

Everything the headers declare, in their order — the details are in each header's part below.

| Name                            | What it does                                                            | Header                 |
|---------------------------------|-------------------------------------------------------------------------|------------------------|
| <code>fk_free</code>            | a buffer the library returned                                           | <code>filekit.h</code> |
| <code>fk_crc32</code>           | (crc: 0 at first, then the last result)                                 | <code>filekit.h</code> |
| <code>fk_adler32</code>         | (adler: 1 at first)                                                     | <code>filekit.h</code> |
| <code>fk_deflate</code>         | The whole of <code>data</code> compressed (level 0..9, -1)              | <code>filekit.h</code> |
| <code>fk_inflate</code>         | ...                                                                     | <code>filekit.h</code> |
| <code>fk_zip_entry</code>       | (a type)                                                                | <code>filekit.h</code> |
| <code>fk_progress</code>        | A step of a long work                                                   | <code>filekit.h</code> |
| <code>fk_zip_open</code>        | 0: err says why                                                         | <code>filekit.h</code> |
| <code>fk_zip_close</code>       | the archive closed, its handle freed (0: nothing)                       | <code>filekit.h</code> |
| <code>fk_zip_error</code>       | the last failure's words                                                | <code>filekit.h</code> |
| <code>fk_zip_password</code>    | for its encrypted entries                                               | <code>filekit.h</code> |
| <code>fk_zip_count</code>       | how many entries it holds (0: no archive)                               | <code>filekit.h</code> |
| <code>fk_zip_find</code>        | its index (letters' case ignored), -1                                   | <code>filekit.h</code> |
| <code>fk_zip_read</code>        | entry <code>i</code> in memory ( <code>fk_free</code> ) → 0 / -1        | <code>filekit.h</code> |
| <code>fk_zip_extract</code>     | entry <code>i</code> as that file → 0 / -1                              | <code>filekit.h</code> |
| <code>fk_zip_extract_all</code> | Everything under prefix (" <code>"</code> or 0                          | <code>filekit.h</code> |
| <code>fk_zipw_create</code>     | nothing written yet; 0: no path                                         | <code>filekit.h</code> |
| <code>fk_zipw_add</code>        | a file, or a folder and all it holds, as <code>name</code> (0: its own) | <code>filekit.h</code> |
| <code>fk_zipw_add_data</code>   | a buffer as an entry (copied)                                           | <code>filekit.h</code> |
| <code>fk_zipw_level</code>      | 0 store, 1 fast, 6 (the default), 9                                     | <code>filekit.h</code> |
| <code>fk_zipw_close</code>      | written → 0 / -1 (err); w freed                                         | <code>filekit.h</code> |
| <code>fk_format</code>          | The formats are asked, not assumed                                      | <code>filekit.h</code> |

| Name                              | What it does                                                     | Header                 |
|-----------------------------------|------------------------------------------------------------------|------------------------|
| <code>fk_arc_formats</code>       | how many there are (out: up to max)                              | <code>filekit.h</code> |
| <code>fk_arc_probe</code>         | What the file at <code>path</code> is                            | <code>filekit.h</code> |
| <code>fk_arc_is_name</code>       | 1: its extension is a format's that is read                      | <code>filekit.h</code> |
| <code>fk_arc_open</code>          | any format read; 0: err says why                                 | <code>filekit.h</code> |
| <code>fk_arc_new</code>           | a new one, not on the card yet ("ZIP"; 0: not a format written)  | <code>filekit.h</code> |
| <code>fk_arc_format</code>        | "ZIP" ...                                                        | <code>filekit.h</code> |
| <code>fk_arc_path</code>          | the archive file's path ("": no archive)                         | <code>filekit.h</code> |
| <code>fk_arc_comment</code>       | the archive's own ("": none)                                     | <code>filekit.h</code> |
| <code>fk_arc_writable</code>      | 1: it can be changed                                             | <code>filekit.h</code> |
| <code>fk_arc_method_name</code>   | entry i: "Deflate", "Store, crypted"...                          | <code>filekit.h</code> |
| <code>fk_arc_test</code>          | entry i read and checked, nothing written → 0 / -1               | <code>filekit.h</code> |
| <code>fk_extract</code>           | (a type)                                                         | <code>filekit.h</code> |
| <code>fk_arc_extract_with</code>  | 0 / -1 ( <code>fk_zip_error</code> )                             | <code>filekit.h</code> |
| <code>fk_arc_extract_bytes</code> | what that selection weighs                                       | <code>filekit.h</code> |
| <code>fk_arc_add</code>           | Changing an archive that can be ( <code>fk_arc_writable</code> ) | <code>filekit.h</code> |
| <code>fk_arc_add_conflicts</code> | names that exist already                                         | <code>filekit.h</code> |
| <code>fk_arc_add_bytes</code>     | the work's size                                                  | <code>filekit.h</code> |
| <code>fk_arc_delete</code>        | Changing an archive that can be ( <code>fk_arc_writable</code> ) | <code>filekit.h</code> |
| <code>fk_arc_rename</code>        | a file, or a folder and what is under it                         | <code>filekit.h</code> |
| <code>fk_arc_new_folder</code>    | Changing an archive that can be ( <code>fk_arc_writable</code> ) | <code>filekit.h</code> |
| <code>fk_arc_write_empty</code>   | a new archive put on the card with nothing in it                 | <code>filekit.h</code> |
| <code>fk_zipmem_count</code>      | its entries, -1: not a ZIP                                       | <code>filekit.h</code> |
| <code>fk_zipmem_entry</code>      | Entry i                                                          | <code>filekit.h</code> |
| <code>fk_zipmem_get</code>        | The entry of that name                                           | <code>filekit.h</code> |

| Name             | What it does                                                                                 | Header    |
|------------------|----------------------------------------------------------------------------------------------|-----------|
|                  | inflated (a 0 byte after its end) → 0 and *out (fk_free), -1                                 |           |
| fk_zipbuf_new    | an empty one; 0: no memory                                                                   | filekit.h |
| fk_zipbuf_add    | 0 / -1                                                                                       | filekit.h |
| fk_zipbuf_finish | the archive (fk_free) → 0 / -1; b freed                                                      | filekit.h |
| fk_zipbuf_free   | (given up before finish)                                                                     | filekit.h |
| fk_exists        | 1 a file, 2 a folder, 0                                                                      | filekit.h |
| fk_file_size     | -1: no such file                                                                             | filekit.h |
| fk_load          | the whole file (a 0 byte after it; fk_free) → 0 / -1                                         | filekit.h |
| fk_save          | 0 / -1                                                                                       | filekit.h |
| fk_mkdirs        | every folder of the path made → 0 / -1                                                       | filekit.h |
| fk_copy          | 0 / -1                                                                                       | filekit.h |
| fk_move          | renamed, else copied then removed                                                            | filekit.h |
| fk_remove        | a file, or a folder and all it holds → 0 / -1                                                | filekit.h |
| fk_tree_size     | The bytes under a path (a file's size                                                        | filekit.h |
| fk_path_name     | after the last '/' or ':' (inside path)                                                      | filekit.h |
| fk_path_ext      | "png" (lower case, without the dot; "": none)                                                | filekit.h |
| fk_path_folder   | "SD:/a/b.txt" → "SD:/a"; "SD:/a" → "SD:/"                                                    | filekit.h |
| fk_path_join     | folder + "/" + name (no '/' added after a '/' or a ':', or before an empty name), cut to cap | filekit.h |
| fk_path_unique   | a name not taken: "a.txt" → "a (2).txt"                                                      | filekit.h |
| fk_human_size    | "12.4 KB"                                                                                    | filekit.h |
| fk_dos_time_str  | "28/09/2026 14:12"                                                                           | filekit.h |
| fs_len           | a string's length in bytes (0 for a null pointer)                                            | fsutil.h  |
| fs_copy          | s into d, cut to cap - 1 bytes, always ended by a 0                                          | fsutil.h  |

| Name                        | What it does                                                                   | Header                |
|-----------------------------|--------------------------------------------------------------------------------|-----------------------|
| <code>fs_lower</code>       | an ASCII capital as its small letter, anything else unchanged                  | <code>fsutil.h</code> |
| <code>fs_ci_cmp</code>      | <code>strcmp</code> with ASCII letters' case ignored: 0 equal, else < 0 or > 0 | <code>fsutil.h</code> |
| <code>fs_join</code>        | <code>out = dir + "/" + name</code> (no doubled slash).                        | <code>fsutil.h</code> |
| <code>fs_basename</code>    | The last path component (points into path).                                    | <code>fsutil.h</code> |
| <code>fs_dirname</code>     | Parent directory of path into out ("SD:/a/b" → "SD:/a", "SD:/a" → "SD:/").     | <code>fsutil.h</code> |
| <code>fs_is_dir</code>      | Parent directory of path into out ("SD:/a/b" → "SD:/a", "SD:/a" → "SD:/").     | <code>fsutil.h</code> |
| <code>fs_exists</code>      | Parent directory of path into out ("SD:/a/b" → "SD:/a", "SD:/a" → "SD:/").     | <code>fsutil.h</code> |
| <code>fs_copy_file</code>   | Parent directory of path into out ("SD:/a/b" → "SD:/a", "SD:/a" → "SD:/").     | <code>fsutil.h</code> |
| <code>fs_skip_dot</code>    | Parent directory of path into out ("SD:/a/b" → "SD:/a", "SD:/a" → "SD:/").     | <code>fsutil.h</code> |
| <code>fs_copy_tree</code>   | Copy a file or a whole folder tree (depth ≤ 12).                               | <code>fsutil.h</code> |
| <code>fs_remove_tree</code> | Delete a file or a whole folder tree (depth ≤ 12).                             | <code>fsutil.h</code> |
| <code>fs_unique_name</code> | A free path in dir for name                                                    | <code>fsutil.h</code> |
| <code>fs_text_fix</code>    | A text file saved by a Windows editor                                          | <code>fsutil.h</code> |
| <code>fk_kv_new</code>      | an empty document (FK_KV_ESCAPES, FK_KV_PIPES); 0: no memory                   | <code>kvtext.h</code> |
| <code>fk_kv_parse</code>    | never fails on syntax (lines without '=')                                      | <code>kvtext.h</code> |

| Name             | What it does                                                                                                                               | Header   |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                  | skipped); 0: no memory                                                                                                                     |          |
| fk_kv_load       | a file read and parsed;<br>0: no such file /<br>unreadable / no<br>memory                                                                  | kvtext.h |
| fk_kv_free       | the document and all its<br>strings (0 accepted)                                                                                           | kvtext.h |
| fk_kv_count      | how many entries                                                                                                                           | kvtext.h |
| fk_kv_key        | entry i's key, trimmed<br>("" out of range)                                                                                                | kvtext.h |
| fk_kv_value      | its value: unescaped, its<br>" " lines joined by '\n'                                                                                      | kvtext.h |
| fk_kv_section    | the name of its section<br>("" before any header)                                                                                          | kvtext.h |
| fk_kv_block      | its block: 0 before any<br>header, then 1, 2...: one<br>per "[...]" line                                                                   | kvtext.h |
| fk_kv_line       | The 1-based line of the<br>value's FIRST line                                                                                              | kvtext.h |
| fk_kv_blocks     | the "[...]" headers read /<br>made                                                                                                         | kvtext.h |
| fk_kv_block_name | block b's name, "level"<br>(b 1-based; "" for 0 / out<br>of range)                                                                         | kvtext.h |
| fk_kv_block_line | its header's line (0:<br>made, not read)                                                                                                   | kvtext.h |
| fk_kv_block_new  | Blocks of the same<br>name, each its own (a<br>list of [alarm]s, a pack's<br>[level]s)                                                     | kvtext.h |
| fk_kv_block_get  | block b's value of key,<br>def if none (b 0 / out of<br>range: def)                                                                        | kvtext.h |
| fk_kv_block_set  | in block b: replaced,<br>else added at its end →<br>0, -1 (no memory, no<br>key, no block b)                                               | kvtext.h |
| fk_kv_get        | The value of the first<br>entry with that section<br>and key, def if none<br>(section "" or 0 = before<br>any header, case-<br>sensitive). | kvtext.h |
| fk_kv_set        | The first match's value<br>replaced, or a new entry<br>at the end of that                                                                  | kvtext.h |

| Name         | What it does                                        | Header   |
|--------------|-----------------------------------------------------|----------|
|              | section's first block → 0, -1 (no memory, no key).  |          |
| fk_kv_remove | the first match removed → 1, 0 none                 | kvtext.h |
| fk_kv_text   | The document as text                                | kvtext.h |
| fk_kv_save   | fk_kv_text written to path (kapi_save_file) → 0, -1 | kvtext.h |

## filekit/filekit.h

filekit.h -- FileKit: what programs do with files, one copy for the whole system (SD:/lib/filekit.so; the shared libraries: docs/03 section 5.6). A program links lib/filekit.imp.a and calls plain C functions (fk\_\*):

```
compression  zlib's deflate and inflate (raw, zlib or gzip wrapped), CRC-32, Adler-32
archives     the formats known asked (fk_arc_formats), an archive of any of them
opened, extracted
              with choices, changed (add, delete, rename) -- ZIP read and written,
tar / tar.gz /
              gzip read
ZIP          an archive on the card read (its entries, one read to memory or
extracted, all of
              them) and made (files and folders of the card, buffers) -- the
Archiver's engine:
              ZIP64, old code pages, ZipCrypto passwords;
              an archive IN MEMORY found into and built (the office formats,
OpenRaster)
files        one loaded whole, saved; a file or a tree copied, moved, removed,
measured; folders made
paths        the name, the extension, the folder of a path; two parts joined; a
size for people
key / value  a text of [section] headers and "key = value" lines read, changed,
written back (kvtext.h)
```

Everything is integer and pointers: a program built without the FPU calls it. A buffer the library returns (void \*\*out) is freed with fk\_free, never with free / delete. The first version: 2026-10-05; the second (the same day): archives of any format, asked from the library (fk\_arc\_\*), tar and gzip read.

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```
void fk_free (void *p);           // a buffer the library returned
```

## compression (zlib)

```
#define FK_RAW      0      // the deflate stream alone (a ZIP entry's)
#define FK_ZLIB    1      // with zlib's header and Adler-32 (PNG, PDF)
#define FK_GZIP    2      // with gzip's header and CRC-32 (.gz, HTTP)
#define FK_AUTO    3      // (inflate only) zlib or gzip, told from the first
bytes
```

```
unsigned fk_crc32 (unsigned crc, const void *data, unsigned n);    // (crc: 0 at
first, then the last result)
unsigned fk_adler32 (unsigned adler, const void *data, unsigned n); // (adler: 1 at
first)
```

The whole of data compressed (level 0..9, -1: the default 6) → 0 and \*out (fk\_free) / \*out\_n, -1.

```
int fk_deflate (const void *data, unsigned n, int wrap, int level, void **out,
unsigned *out_n);
```

... and back. size\_hint: the size expected (0: not known; the buffer grows) → 0 / -1 (damaged).  
The result has a 0 byte after its end (a text can be used as it is).

```
int fk_inflate (const void *data, unsigned n, int wrap, unsigned size_hint, void
**out, unsigned *out_n);
```

## a ZIP archive on the card

```
struct fk_zip_entry
{
    char      name[300];      // its path in the archive (UTF-8, '/' between
the parts)
    unsigned long long size, packed;    // the original size, what it takes in the
archive
    unsigned   crc, dos_time;    // CRC-32; the DOS date and time
(fk_dos_time_str)
    int        dir, encrypted;    // a folder; needs the password
    int        method;           // 0 stored, 8 deflate...
    int        reserved[4];
};
typedef struct fk_zip fk_zip;
```

A step of a long work: done of total bytes, the file being worked on (0: the same) → 0 to go on,  
anything else to stop.

```
typedef int (*fk_progress) (void *user, unsigned long long done, unsigned long long
total, const char *name);
```

```
fk_zip *fk_zip_open (const char *path, char *err, int cap);    // 0: err says why
void fk_zip_close (fk_zip *z);    // the archive closed, its handle freed (0: nothing)
const char *fk_zip_error (fk_zip *z);    // the last failure's words
void fk_zip_password (fk_zip *z, const char *password);    // for its encrypted
entries
int fk_zip_count (fk_zip *z);    // how many entries it holds (0: no archive)
int fk_zip_entry (fk_zip *z, int i, struct fk_zip_entry *out);    // 1 / 0
int fk_zip_find (fk_zip *z, const char *name);    // its index (letters'
case ignored), -1
int fk_zip_read (fk_zip *z, int i, void **out, unsigned *out_n);    // entry i in
memory (fk_free) -> 0 / -1
int fk_zip_extract (fk_zip *z, int i, const char *dest_path);    // entry i as
that file -> 0 / -1
```

Everything under `prefix` ("" or 0: the whole archive) into the folder `dest_dir`, the folders made, the names made safe for the card → the files written, -1 (`fk_zip_error`; stopped: what was done stays).

```
int fk_zip_extract_all (fk_zip *z, const char *prefix, const char *dest_dir,
fk_progress cb, void *user);
```

A new archive (an existing file is replaced when it is closed): its entries said first, written by `fk_zipw_close`.

```
typedef struct fk_zipw fk_zipw;
fk_zipw *fk_zipw_create (const char *path); // nothing written yet; 0: no path
int fk_zipw_add (fk_zipw *w, const char *disk_path, const char *name); // a file, or
a folder and all it holds, as `name` (0: its own)
int fk_zipw_add_data (fk_zipw *w, const char *name, const void *data, unsigned n);
// a buffer as an entry (copied)
void fk_zipw_level (fk_zipw *w, int level); // 0 store, 1 fast, 6 (the
default), 9
int fk_zipw_close (fk_zipw *w, fk_progress cb, void *user, char *err, int cap); //
written -> 0 / -1 (err); w freed
```

## archives of any format the library knows (version 2, 2026-10-05)

The formats are asked, not assumed: a program that shows archives (the Archiver) lists them from here, so a format added to the library is one it has at once.

```
struct fk_format
{
    char name[16]; // "ZIP", "TAR", "TAR.GZ", "GZIP"
    char extensions[96]; // "zip jar docx ..." (lower case, a space
between)
    char note[80]; // what can be done with it, in a few words
    int can_read, can_write; // opened and extracted; made and changed (add,
delete, rename)
    int can_password; // encrypted entries are read (with fk_arc_password)
    int reserved[4];
};
int fk_arc_formats (struct fk_format *out, int max); // how many there are
(out: up to max)
```

What the file at `path` is: 1 and its format's name -- or 0 and why ("7z archives are not supported yet.").

```
int fk_arc_probe (const char *path, char *format, int fcap, char *why, int wcap);
int fk_arc_is_name (const char *name); // 1: its extension is a
format's that is read
```

An archive opened: the same handle as `fk_zip` (`fk_zip_count`, `_entry`, `_find`, `_read`, `_extract`, `_extract_all`, `_password`, `_error` and `_close` work on it whatever its format).

```
typedef struct fk_zip fk_arc;
fk_arc *fk_arc_open (const char *path, char *err, int cap); // any format read;
0: err says why
fk_arc *fk_arc_new (const char *path, const char *format); // a new one, not on
the card yet ("ZIP"; 0: not a format written)
const char *fk_arc_format (fk_arc *a); // "ZIP" ...
const char *fk_arc_path (fk_arc *a); // the archive file's path ("": no archive)
const char *fk_arc_comment (fk_arc *a); // the archive's own ("":
none)
int fk_arc_writable (fk_arc *a); // 1: it can be changed
int fk_arc_method_name (fk_arc *a, int i, char *out, int cap); // entry i:
```

```
"Deflate", "Store, crypted"...
int fk_arc_test (fk_arc *a, int i, fk_progress cb, void *user); // entry i read and
checked, nothing written -> 0 / -1
```

Extracting with choices.

```
#define FK_LAYOUT_FULL      0      // the archive's folders kept
#define FK_LAYOUT_FROM      1      // ... from the folder `from` down (what is above
it dropped)
#define FK_LAYOUT_FLAT      2      // the files alone, no folder
#define FK_EXISTS_ASK       0      // a file that exists: ask (the callback),
replace, skip, keep both
#define FK_EXISTS_REPLACE   1
#define FK_EXISTS_SKIP      2
#define FK_EXISTS_KEEP_BOTH 3
#define FK_ANSWER_REPLACE   1      // the callback's answers
#define FK_ANSWER_SKIP      2
#define FK_ANSWER_KEEP_BOTH 3
#define FK_ANSWER_REPLACE_ALL 4
#define FK_ANSWER_SKIP_ALL  5
#define FK_ANSWER_CANCEL    (-1)
struct fk_extract
{
    unsigned    size;           // sizeof (struct fk_extract)
    const char  *dest;          // the folder to extract into (made)
    const char  *selected;      // one byte an entry, not 0: taken (0: every
entry)
    const char  *from;          // FK_LAYOUT_FROM: the archive's folder shown
    int         layout, exists;
    int         (*ask) (void *user, const char *path); // FK_EXISTS_ASK -> an
FK_ANSWER_*
    fk_progress progress;
    void        *user;          // (both callbacks')
    int         files, skipped;  // out: what was done
};
int fk_arc_extract_with (fk_arc *a, struct fk_extract *x); // 0 / -1
(fk_zip_error)
unsigned long long fk_arc_extract_bytes (fk_arc *a, const char *selected); // what
that selection weighs
```

Changing an archive that can be (fk\_arc\_writable): it is rewritten beside itself, then swapped in, and its entries are read again (the indexes change). → 0 / -1 (fk\_zip\_error). paths: files and folders of the card, put in the archive's folder into ("": its top); keep\_folders: a folder keeps its name and tree; replace: a name that exists is replaced, else skipped.

```
int fk_arc_add (fk_arc *a, const char *const *paths, int n, const char *into, int
keep_folders, int replace, int level,
    fk_progress cb, void *user, int *added);
int fk_arc_add_conflicts (fk_arc *a, const char *const *paths, int n, const char
*into, int keep_folders); // names that exist already
unsigned long long fk_arc_add_bytes (fk_arc *a, const char *const *paths, int n,
const char *into, int keep_folders, int replace); // the work's size
int fk_arc_delete (fk_arc *a, const char *selected, fk_progress cb, void *user);
int fk_arc_rename (fk_arc *a, const char *from, const char *to, fk_progress cb, void
*user); // a file, or a folder and what is under it
int fk_arc_new_folder (fk_arc *a, const char *name, fk_progress cb, void *user);
int fk_arc_write_empty (fk_arc *a); // a new archive put on the card
with nothing in it
```

## a ZIP archive in memory (a document read whole: .docx, .xlsx, .odt, .ora ...)

```
int fk_zipmem_count (const void *zip, unsigned n);           // its entries, -1: not a ZIP
```

Entry i: its name (cap bytes) and its size → 1 / 0.

```
int fk_zipmem_entry (const void *zip, unsigned n, int i, char *name, int cap, unsigned *size);
```

The entry of that name inflated (a 0 byte after its end) → 0 and \*out (fk\_free), -1: none, damaged.

```
int fk_zipmem_get (const void *zip, unsigned n, const char *name, void **out, unsigned *out_n);
```

... and one built: entries added (level 0: stored -- the "mimetype" of an OpenDocument first), then the archive's bytes.

```
typedef struct fk_zipbuf fk_zipbuf;
fk_zipbuf *fk_zipbuf_new (void);           // an empty one; 0: no memory
int fk_zipbuf_add (fk_zipbuf *b, const char *name, const void *data, unsigned n, int level); // 0 / -1
int fk_zipbuf_finish (fk_zipbuf *b, void **out, unsigned *out_n); // the archive (fk_free) -> 0 / -1; b freed
void fk_zipbuf_free (fk_zipbuf *b);        // (given up before finish)
```

## files

```
int fk_exists (const char *path);           // 1 a file, 2 a folder, 0
long long fk_file_size (const char *path); // -1: no such file
int fk_load (const char *path, void **out, unsigned *out_n); // the whole file (a 0 byte after it; fk_free) -> 0 / -1
int fk_save (const char *path, const void *data, unsigned n); // 0 / -1
int fk_mkdirs (const char *path);           // every folder of the path made -> 0 / -1
```

A file, or a folder and all it holds. dst is the new path itself (not the folder to put it in).

```
int fk_copy (const char *src, const char *dst, fk_progress cb, void *user); // 0 / -1
int fk_move (const char *src, const char *dst, fk_progress cb, void *user); // renamed, else copied then removed
int fk_remove (const char *path);           // a file, or a folder and all it holds -> 0 / -1
```

The bytes under a path (a file's size; a folder's whole tree); \*files, \*folders: how many (0: not wanted).

```
long long fk_tree_size (const char *path, int *files, int *folders);
```

## paths

```
const char *fk_path_name (const char *path); // after the last '/' or ':' (inside path)
void fk_path_ext (const char *path, char *out, int cap); // "png" (lower case, without the dot; "": none)
void fk_path_folder (const char *path, char *out, int cap); // "SD:/a/b.txt" -> "SD:/a"; "SD:/a" -> "SD:/"
void fk_path_join (char *out, int cap, const char *folder, const char *name); // folder + "/" + name (no '/' added after a '/' or a ':', or before an empty name), cut to cap
void fk_path_unique (char *path, int cap); // a name not taken: "a.txt" -> "a (2).txt"
```

```
void fk_human_size (unsigned long long bytes, char *out, int cap); // "12.4 KB"
void fk_dos_time_str (unsigned dos_time, char *out, int cap); // "28/09/2026
14:12"

}
#endif
```

(the file helpers -- fs\_join, fs\_exists, fs\_copy\_tree...: fsutil.h, C++)

(a text document of [section] headers and "key = value" lines, read and written back -- fk\_kv\_\*: kvtext.h, C and C++)

## filekit/fsutil.h

fsutil.h -- file-system helpers shared by the file tools (File Viewer, trash, dock): path join, exists / is\_dir, recursive copy and delete, a free "name copy" name. FileKit's (filekit.so): this header declares, fsutil.inc has the code (it was user/fsutil.h, a header of inline functions, until 2026-10-05). C++.

(a program for Onyx: FileKit's functions, by name. A PC build, or FS\_INLINE: the code inline in the program -- no shared library there.)

```
#define FS_API extern "C"
#define FS_API static inline

#define FS_NAMELEN 72
#define FS_PATHLEN 300
int fs_len (const char *s); // a string's length in bytes (0 for a null pointer)
void fs_copy (char *d, const char *s, int cap); // s into d, cut to cap - 1 bytes,
always ended by a 0
char fs_lower (char c); // an ASCII capital as its small letter, anything else
unchanged
int fs_ci_cmp (const char *a, const char *b); // strcmp with ASCII letters' case
ignored: 0 equal, else < 0 or > 0
```

out = dir + "/" + name (no doubled slash).

```
void fs_join (char *out, int cap, const char *dir, const char *name);
```

The last path component (points into path).

```
const char *fs_basename (const char *path);
```

Parent directory of path into out ("SD:/a/b" → "SD:/a", "SD:/a" → "SD:/").

```
void fs_dirname (char *out, int cap, const char *path);
bool fs_is_dir (const char *path);
bool fs_exists (const char *path);
bool fs_copy_file (const char *src, const char *dst);
bool fs_skip_dot (const char *n);
```

Copy a file or a whole folder tree (depth ≤ 12).

```
bool fs_copy_tree (const char *src, const char *dst, int depth = 0);
```

Delete a file or a whole folder tree (depth ≤ 12).

```
bool fs_remove_tree (const char *path, int depth = 0);
```

A free path in dir for name: "name", then "name copy", "name copy 2", ... (the extension stays at the end: "notes copy.txt").

```
void fs_unique_name (char *out, int cap, const char *dir, const char *name, const
char *tag = " copy");
```

A text file saved by a Windows editor: drop a UTF-8 BOM, turn UTF-16 (Notepad) into 8-bit. In place (b[n] must be writable); returns the new length.

```
int fs_text_fix (char *b, int n);
```

## filekit/kvtext.h

kvtext.h -- FileKit: a text document of [section] headers and "key = value" lines, read and written back (a game's progress, a level pack, a settings file bigger than AppKit's .ini reader takes). The entries are kept in the file's order; a section may come back many times (a pack's [level] blocks: each "[...]" line is a block of its own); a value may hold new lines -- written "\n" (FK\_KV\_ESCAPES: a progress file) or as the lines that follow, each starting with "|" (FK\_KV\_PIPES: a pack's tables and solutions); the line of every value is kept (an error message's "line 42"). Unknown keys are kept: a program writes back what it read with only its own keys changed. No limit on sizes but memory. C and C++.

```
[pack]                                '#' and ';' start a comment line; a blank line or a
comment ends a "|" block
title = 1. Gates
[level]                                fk_kv_block: 2 (the second "[...]" line);
fk_kv_section: "level"
table =                                fk_kv_value: "A | Out\n0 | 0\n1 | 1"; fk_kv_line: 5
(its first "|" line),
| A | Out                                the value's line j at fk_kv_line + j
| 0 | 0
| 1 | 1
```

FileKit's (filekit.so) since 2026-10-06 (AutoDev round 2, Circuits): this header declares, kvtext.inc has the code (on the PC the code is inline in the program, as fsutil.h's).

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(a program for Onyx: FileKit's functions, by name. A PC build, or FK\_KV\_INLINE: the code inline in the program -- no shared library there. FK\_KV\_IMPL: the library's own build, kvtext.cpp.)

```
#   define FK_KV_API   extern "C"
#   define FK_KV_API   extern
#   define FK_KV_API static inline
```

```
typedef struct fk_kv fk_kv;           // a document (its entries, its blocks), made by
fk_kv_new / parse / load
#define FK_KV_ESCAPES 1               // values may hold new lines and '\': written "\n", "\\n", read back (progress.ini)
#define FK_KV_PIPES 2                // a value may go on in the lines that follow, each
starting with "|" (packs)
```

## making and freeing

```
fk_kv *fk_kv_new (int flags); // an empty document (FK_KV_ESCAPES, FK_KV_PIPES); 0: no memory
fk_kv *fk_kv_parse (const char *text, int flags); // never fails on syntax (lines without '=' skipped); 0: no memory
fk_kv *fk_kv_load (const char *path, int flags); // a file read and parsed; 0: no such file / unreadable / no memory
void fk_kv_free (fk_kv *kv); // the document and all its strings (0 accepted)
```

## the entries, in the file's order

```
int fk_kv_count (const fk_kv *kv); // how many entries
const char *fk_kv_key (const fk_kv *kv, int i); // entry i's key, trimmed (" out of range)
const char *fk_kv_value (const fk_kv *kv, int i); // its value: unescaped, its "|" lines joined by '\n'
const char *fk_kv_section (const fk_kv *kv, int i); // the name of its section (" before any header)
int fk_kv_block (const fk_kv *kv, int i); // its block: 0 before any header, then 1, 2...: one per "[...]" line
```

The 1-based line of the value's FIRST line: the key's line, or its first "|" line when "key =" is empty; the value's line j is at fk\_kv\_line + j. 0: an entry made by fk\_kv\_set, not read.

```
int fk_kv_line (const fk_kv *kv, int i);
```

## the blocks (one per "[...]" line)

```
int fk_kv_blocks (const fk_kv *kv); // the "[...]" headers read / made
const char *fk_kv_block_name (const fk_kv *kv, int b); // block b's name, "level" (b 1-based; "" for 0 / out of range)
int fk_kv_block_line (const fk_kv *kv, int b); // its header's line (0: made, not read)
```

Blocks of the same name, each its own (a list of [alarm]s, a pack's [level]s): a new "[name]" block made at the end → its number (1-based), -1 no memory. fk\_kv\_set reaches only a name's first block; these, any block by number.

```
int fk_kv_block_new (fk_kv *kv, const char *name);
const char *fk_kv_block_get (const fk_kv *kv, int b, const char *key, const char *def); // block b's value of key, def if none (b 0 / out of range: def)
int fk_kv_block_set (fk_kv *kv, int b, const char *key, const char *value); // in block b: replaced, else added at its end -> 0, -1 (no memory, no key, no block b)
```

## looking up and changing

The value of the first entry with that section and key, def if none (section "" or 0 = before any header, case-sensitive).

```
const char *fk_kv_get (const fk_kv *kv, const char *section, const char *key, const char *def);
```

The first match's value replaced, or a new entry at the end of that section's first block → 0, -1 (no memory, no key). Section "" or 0 = before the first header; no block of that name = a new block at the end.

```
int fk_kv_set (fk_kv *kv, const char *section, const char *key, const char *value);
int fk_kv_remove (fk_kv *kv, const char *section, const char *key); // the first match removed -> 1, 0 none
```

## writing

The document as text: the comment as a first line ("# " put before it unless it starts with '#'; 0: none), the entries before any header, then each block "[name]" with its entries (ESCAPES: values escaped; PIPES: a multi-line value as "key =" and its "|" lines; neither: its new lines written as spaces). Written then parsed with the same flags, it gives the same entries. The pointer is the document's, valid until its next change / free; \*len its length (0 accepted).

```
const char *fk_kv_text (fk_kv *kv, const char *comment, int *len);  
int fk_kv_save (fk_kv *kv, const char *path, const char *comment); // fk_kv_text  
written to path (kapi_save_file) -> 0, -1
```